

CSE 225: Data Structures and Algorithms

Course Outline

Module 1: Basic Data Structures

- Array and Linked List
- Stacks and Queues
- Trees

Module 2: Dynamic Array and Amortized Analysis

Module 3: Priority Queues and Disjoint Sets

Module 4: Hash Tables

Module 5: Binary Search Trees

Module 6: Graphs

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Course Outcomes(COs)

Upon Successful completion of this course, students will be able to:

Sl.	CO Description	Weightage (%)
1	Understand the fundamental Data Structures including arrays, linked lists, trees, binary search trees, stacks, queues, priority queues, graphs, and hash tables.	70%
2	Identify appropriate data structures based on algorithmic complexity for solving real-world problems	10%
3	Use programming tools for the implementation of abstract data types (ADT)	20%

Introduction to Algorithm

(let's discuss this before we jump into our discussion of the data structure)

What is an algorithm?

- Set of well defined rules, a recipe in effect for solving some computational problem.
- Example 1: Maybe you have a roadmap and an origin and a destination and you want to compute the shortest path from that origin to that destination
- Example 2: May be you face a number of different tasks that need to be completed by certain deadlines and you ant to know in what order you should accomplish the task. So that you complete them all by their respective deadlines.

Let's design an algorithm and do the analysis

Case I: We will do integer Multiplication

Integer Multiplication

Input : 2 'n' digit numbers x and y

Output : product $x*y$

The High School Algorithm

A handwritten multiplication problem showing the high school algorithm. The multiplicand is 5678 and the multiplier is 1234. The partial products are 5678, 11356, 56780, and 567800. The final product is 7006652. A pink arrow on the left indicates the number of rows (n) is 5. A pink oval highlights the first row of the partial products (5678).

$$\begin{array}{r} 5678 \\ \times 1234 \\ \hline 5678 \\ 11356 \\ 56780 \\ 567800 \\ \hline 7006652 \end{array}$$

n rows

n

of operations $\sim \text{constant} * n^2$

CAN WE DO BETTER?

Let's design an algorithm and
do the analysis

Case II: We will do merge sort

The Sorting Problem

Input : array of n numbers, unsorted.

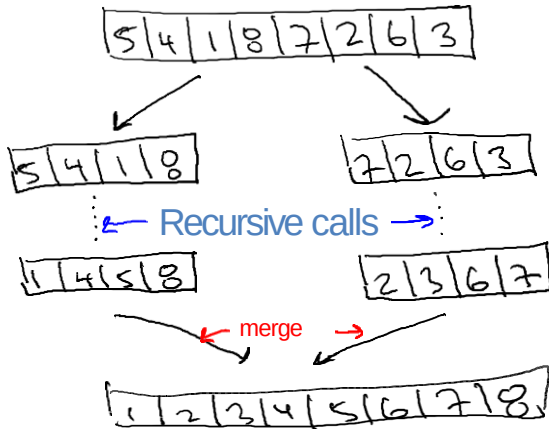
5 4 1 8 7 2 6 3

Assume
Distinct

Output : Same numbers, sorted in increasing order

1 2 3 4 5 6 7 8

Merge Sort: Example



Merge Sort: Pseudocode

- recursively sort 1st half of the input array
- recursively sort 2nd half of the input array
- merge two sorted sublists into one

Pseudocode for Merge:

C = output [length = n]
A = 1st sorted array [n/2]
B = 2nd sorted array [n/2]
i = 1
j = 1

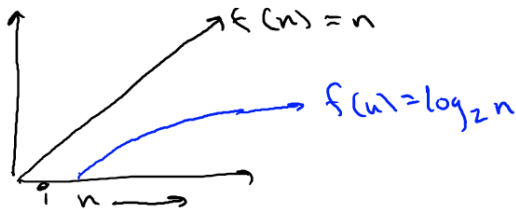
for k = 1 to n
 if A(i) < B(j)
 C(k) = A(i)
 i++
 else [B(j) < A(i)]
 C(k) = B(j)
 j++
end

(ignores end cases)

Running Time of Merge Sort

Claim : Merge Sort requires
 $\leq 6n \log_2 n + 6n$ operations
to sort n numbers.

Recall : $\log_2 n$ is the #
of times you divide by 2
until you get down to 1



Next Class: Asymptotic Analysis
“big-oh” notation